ME595 Project Proposal: SINDy-RL for Inverted Double Pendulum Control

Andrew Falcone & Patrick Smith

Introduction

Reinforcement learning (RL) is a powerful approach to control in which an agent learns a policy by interacting with a simulated environment and receiving feedback in the form of a reward signal. A primary challenge of RL is that training may require a significant number of simulation interactions before learning an effective control policy. Our project plans to study whether a learned dynamics model can reduce the computational cost of training for the double inverted pendulum problem, a standard non-linear control benchmark in which a cart must balance two linked pendulum arms upright.

We plan to use Sparse Identification of Nonlinear Dynamics with control (SINDYc) to learn a surrogate model of the system dynamics from state-action trajectory data. SINDY identifies governing equations by selecting a sparse selection of candidate functions, such as polynomials, to determine the ones that best describe how the system evolves over time. The model learns a mapping from the current state and action (x_k, u_k) to the next state x_{k+1}. This model can then be used to replace the original simulator as part of the training process. The surrogate model replaces the full physics engine by learning a simplified mapping from state and action to next state instead of evaluating a full physics model, which is significantly cheaper to compute. This approach is inspired by the SINDy-RL framework proposed by Zolman et al. [1] which demonstrated that using learned surrogate models can greatly reduce the number of required environment interactions.

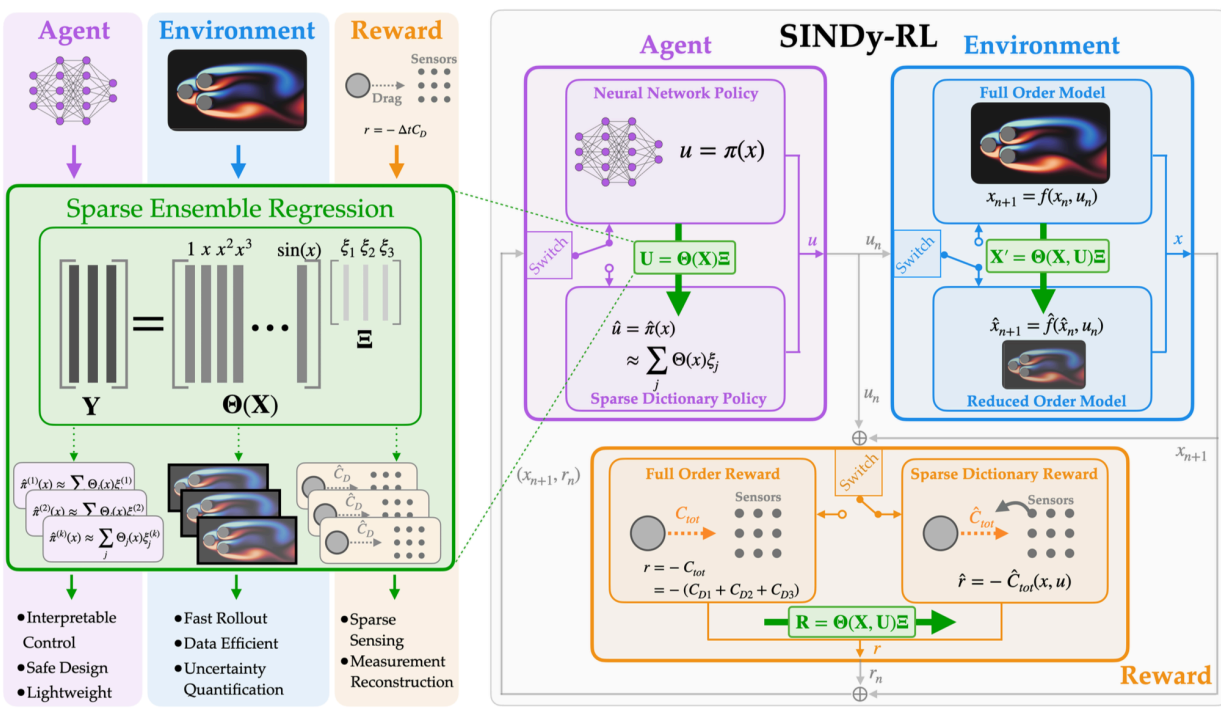

Figure 1: The SINDy-RL training loop. An E-SINDy surrogate is fit from trajectory data collected in the full-order environment. An RL policy is then trained primarily inside the surrogate, with periodic returns to the real environment to collect new data and refresh the model. Adapted from Zolman et al. (2024) [1].

Primary Goals

Our primary two goals are to first learn a controlled dynamics model from simulator data through Gymnasium/MuJoCo's *InvertedDoublePendulum-v5* environment [3]. The action is a one dimensional force applied to the cart and the observations include a 9 dimensional vector including cart position, sin and cosine encodings of pole angles, the velocities of position and angles, and a constraint force of the cart. All training data is generated through agent-environment interaction; no external datasets are required. As an optional extension to improve state space coverage, we may augment our dataset with the Minari training set "mujoco/inverteddoublependulum/expert-v0". The input to our model will be the current 9D state x_k and action u_k and output will be next state x_{k+1}. We will fit SINDY models using PySINDy [4] starting with polynomial feature libraries. Key evaluation metrics for the surrogate include one-step prediction error, multi-step rollout error, and whether the surrogate captures qualitatively correct behavior near the unstable upright equilibrium in the original simulator environment.

Our second goal is to compare control performance using reinforcement learning. We will train a baseline policy directly in the Gymnasium/MuJoCo environment using Proximal Policy Optimization (PPO), a standard RL algorithm for continuous controls. We will then test whether the policy trained using the SINDy surrogate and evaluated in the original environment compares with the one trained on the full environment with fewer simulator interactions. Success will be measured using episode reward, sample efficiency (number of simulator interactions required), and stability of learned policies in the original simulator environment.

Backup Goal

If we run into issues training the SINDy-RL controller, we will either focus on modeling the dynamics with SINDy, or train the RL controller using the full order simulator, depending on the obstacle.

Stretch Goal

If the primary goal is completed with time remaining, we will extend the surrogate modeling comparison to include Dynamic Mode Decomposition with controls (DMDc) as a linear data-driven alternative to SINDy and compare sample efficiency and final performance against both the SINDy surrogate and full-order baselines.

Division of Responsibilities

Andrew Falcone will be responsible for simulator data collection, state representation choices, SINDy model fitting, and surrogate validation. This includes writing code to collect trajectories, fitting controlled dynamics models, tuning sparse regression settings, and evaluating both one-step and multi-step prediction errors.

Patrick Smith will be responsible for control policy synthesis including PPO baseline training on the full-order environment and the SINDy surrogate as well as evaluating sample efficiency and model performance.

Both team members share responsibility for the common evaluation pipeline, final results analysis, and written report.

CRediT Statement

Andrew Falcone: Conceptualization, Writing — original draft (Introduction, surrogate dynamics modeling sections), Writing — review and editing (control policy sections).

Patrick Smith: Conceptualization, Writing — original draft (control policy sections, backup and stretch goals), Writing — review and editing (surrogate dynamics modeling sections).

References

[1] N. Zolman, C. Lagemann, U. Fasel, J. N. Kutz, and S. L. Brunton, "SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning," *Nature Communications*, 2025. arXiv:2403.09110.

[2] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *PNAS*, 113(15):3932-3937, 2016.

[3] M. Towers, J. K. Terry, A. Kwiatkowski et al., "Gymnasium," arXiv:2407.17032, 2024.

[4] B. M. de Silva, K. Champion, M. Quade, J.-C. Loiseau, J. N. Kutz, and S. L. Brunton, "PySINDy: A Python package for the sparse identification of nonlinear dynamical systems from data," *JOSS*, 5(49):2104, 2020.